

Oscar Projekt

SAPunanje mačora

2025-11-02

Oscar Projekt

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(readr)
library(stringr)
library(hflights)
library(tidyr)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
library(ggplot2)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1      v tibble 3.3.0
## v purrr 1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggpubr)
```

```
## Warning: package 'ggpubr' was built under R version 4.5.2
```

```
library(boot)
library(gridExtra)
```

```
## Warning: package 'gridExtra' was built under R version 4.5.2
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##      combine
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.5.2
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.5.2
```

```
##
## Attaching package: 'car'
##
## The following object is masked from 'package:boot':
##
##      logit
##
## The following object is masked from 'package:purrr':
##
##      some
##
## The following object is masked from 'package:dplyr':
##
##      recode
```

```
oscars_dataset <- read_delim("oscars_dataset.csv", delim = ";", escape_double = FALSE, trim_ws = TRUE)
```

```
## Rows: 571 Columns: 20
## -- Column specification -----
## Delimiter: ";"
## chr   (10): Film, Oscar Year, Film Studio/Producer(s), Award, Movie Genre, Ma...
## dbl   (8): Unnamed: 0, Year of Release, Movie Time, IMDB Rating, Tomatometer...
## num   (1): IMDB Votes
## date  (1): Original Release Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Including Plots

You can also embed plots, for example:

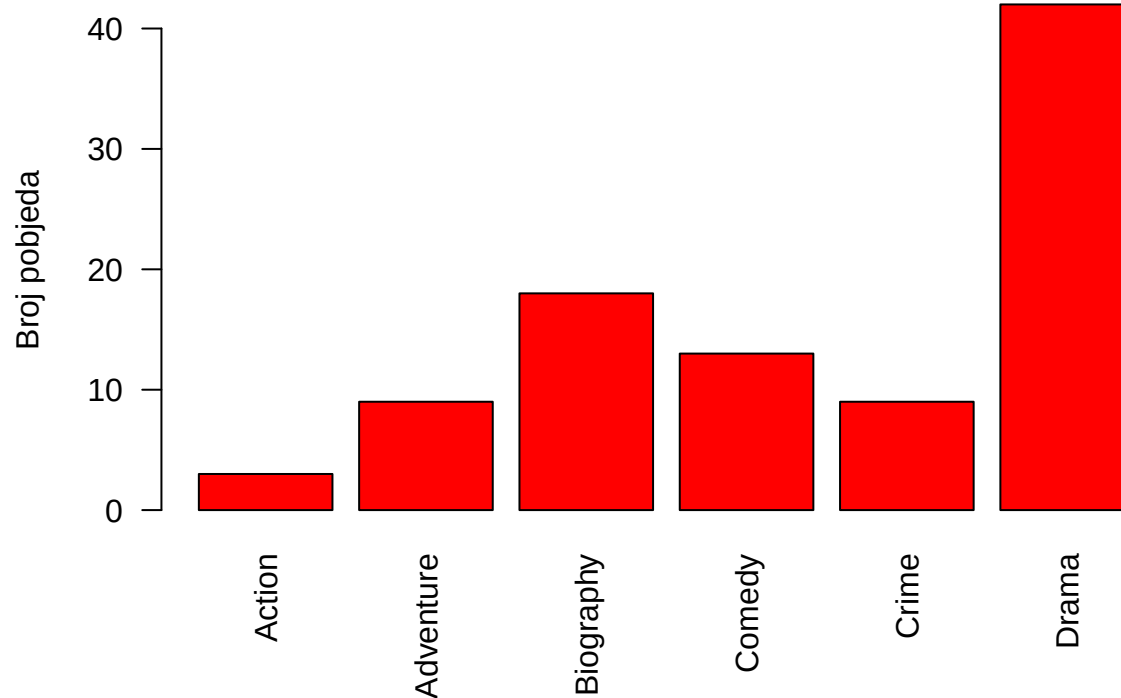
```
##      Unnamed: 0      Film      Oscar Year      Film Studio/Producer(s)
##  Min.   : 0.0      Length:571      Length:571      Length:571
## 1st Qu.:142.5      Class :character      Class :character      Class :character
## Median :285.0      Mode  :character      Mode  :character      Mode  :character
## Mean   :285.0
## 3rd Qu.:427.5
## Max.   :570.0
##
##      Award      Year of Release      Movie Time      Movie Genre
## Length:571      Min.   :1927      Min.   : 66.0      Length:571
## Class :character      1st Qu.:1944      1st Qu.:107.0      Class :character
## Mode  :character      Median :1972      Median :121.0      Mode  :character
##                               Mean  :1973      Mean  :124.9
##                               3rd Qu.:2001      3rd Qu.:136.5
##                               Max.   :2021      Max.   :238.0
##
##      Main Genre      Subgenre      IMDB Rating      IMDB Votes
## Length:571      Length:571      Min.   :5.60      Min.   : 18
## Class :character      Class :character      1st Qu.:7.30      1st Qu.: 9660
## Mode  :character      Mode  :character      Median :7.60      Median : 58996
##                               Mean  :7.57      Mean  : 208480
##                               3rd Qu.:7.90      3rd Qu.: 259429
##                               Max.   :9.30      Max.   :2452594
##
##      Content Rating      Directors      Original Release Date      Production Company
## Length:571      Length:571      Min.   :1927-08-12      Length:571
## Class :character      Class :character      1st Qu.:1951-11-12      Class :character
## Mode  :character      Mode  :character      Median :1979-01-01      Mode  :character
##                               Mean  :1978-05-30
##                               3rd Qu.:2006-08-31
##                               Max.   :2021-02-26
##                               NA's   :112
##      Tomatometer Rating      Tomatometer Count      Audience Rating      Audience Count
## Min.   : 29.00      Min.   : 5.0      Min.   :21.00      Min.   : 6
## 1st Qu.: 84.00      1st Qu.: 26.0      1st Qu.:78.00      1st Qu.: 6270
## Median : 91.00      Median : 59.0      Median :85.00      Median : 35290
## Mean   : 87.64      Mean   :124.1      Mean   :82.57      Mean   : 493614
## 3rd Qu.: 95.00      3rd Qu.:217.0      3rd Qu.:90.00      3rd Qu.: 113078
## Max.   :100.00      Max.   :598.0      Max.   :98.00      Max.   :35797635
## NA's   :112      NA's   :112      NA's   :112      NA's   :112
```

#Jesu li među pobjednicima neki žanrovi češći? Hipoteze:

H0 - Svi žanrovi imaju jednaku vjerojatnost pobjede H1 - Postoji razlika u učestalosti pobjeda među žanrovima

alpha = 0.05

Broj pobjednika po žanru



```
## [1] "Kontingencijska tablica (frekvencije):"
```

```
##
##      Nominee Winner
## Action      36     3
## Adventure   36     9
## Animation    3     0
## Biography   83    18
## Comedy     95    13
## Crime       39     9
## Drama      173    42
## Family       1     0
## Film-Noir    2     0
## Horror       3     0
## Music        1     0
## Musical      2     0
## Romance      1     0
## Western      2     0
```

```
## [1] "Kontingencijska tablica za subžanrove (frekvencije):"
```

```
##
##      Nominee Winner
## Adventure    20     2
## Biography     6     2
```

```
## Comdey      1      0
## Comedy      4      2
## Crime       13     3
## Drama      215    40
## Family      11     4
## Fantasy      8     0
## History     12     1
## Musical     20     3
## Mystery     13     1
## Noir        2      2
## None        43    11
## Romance     77    15
## Sci-Fi       4     0
## Sport        2     2
## Thriller     8     1
## War         14     3
## Western      4     2
```

```
## Warning in chisq.test(tablica_zanr_award): Chi-squared approximation may be
## incorrect
```

```
## [1] "Očekivane frekvencije:"
```

```
##
##      Nominee      Winner
## Action  32.5796848  6.4203152
## Adventure 37.5919440  7.4080560
## Animation  2.5061296  0.4938704
## Biography 84.3730298 16.6269702
## Comedy   90.2206655 17.7793345
## Crime    40.0980736  7.9019264
## Drama   179.6059545 35.3940455
## Family    0.8353765  0.1646235
## Film-Noir 1.6707531  0.3292469
## Horror    2.5061296  0.4938704
## Music     0.8353765  0.1646235
## Musical   1.6707531  0.3292469
## Romance   0.8353765  0.1646235
## Western   1.6707531  0.3292469
```

```
## Warning in chisq.test(tablica_zanr_award2): Chi-squared approximation may be
## incorrect
```

```
## [1] "Očekivane frekvencije za subžanrove:"
```

```
##
##      Nominee      Winner
## Adventure 18.3782837  3.6217163
## Biography  6.6830123  1.3169877
## Comdey     0.8353765  0.1646235
## Comedy     5.0122592  0.9877408
## Crime     13.3660245  2.6339755
```

```
## Drama      213.0210158 41.9789842
## Family     12.5306480  2.4693520
## Fantasy     6.6830123  1.3169877
## History    10.8598949  2.1401051
## Musical    19.2136602  3.7863398
## Mystery    11.6952715  2.3047285
## Noir       3.3415061  0.6584939
## None      45.1103327  8.8896673
## Romance    76.8546410 15.1453590
## Sci-Fi     3.3415061  0.6584939
## Sport      3.3415061  0.6584939
## Thriller   7.5183888  1.4816112
## War       14.2014011  2.7985989
## Western    5.0122592  0.9877408
```

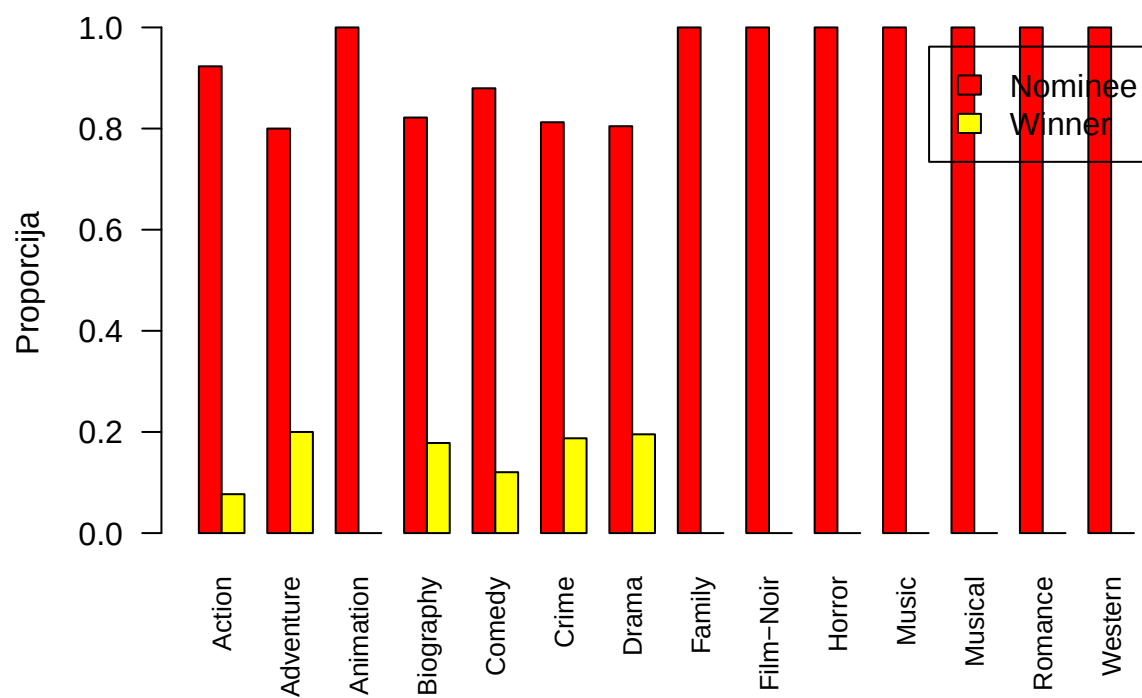
```
## [1] "Rezultat testa za genre: "
```

```
##
## Pearson's Chi-squared test
##
## data:  tablica_zanr_award
## X-squared = 8.8789, df = 13, p-value = 0.782
```

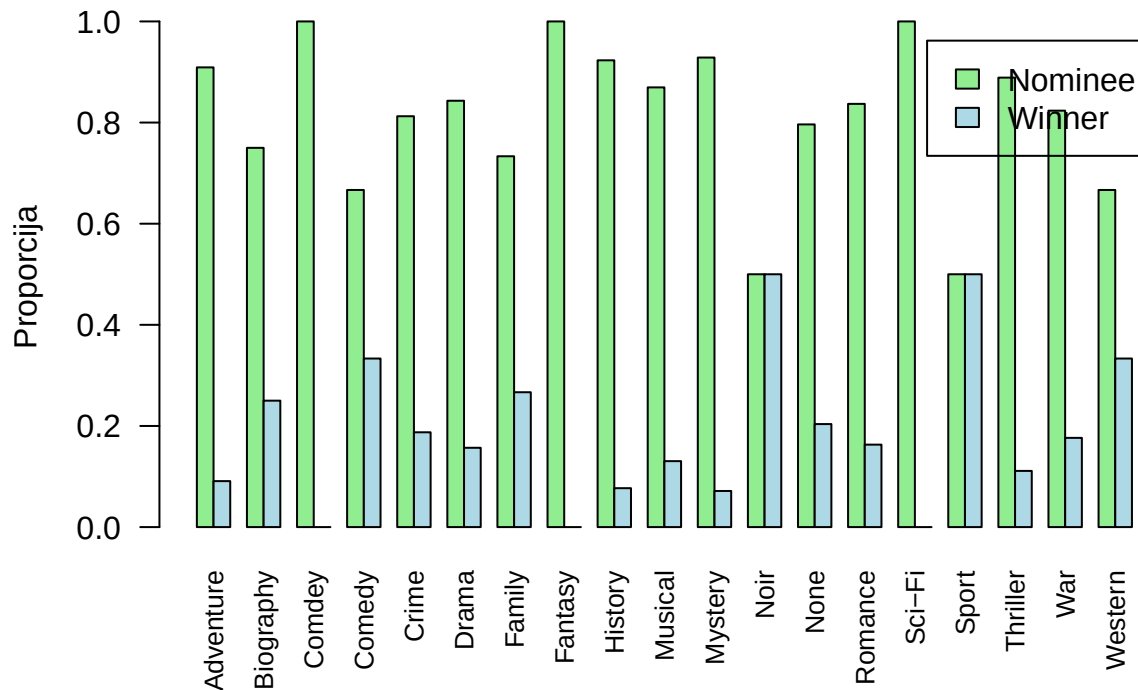
```
## [1] "Rezultat testa za subgenre: "
```

```
##
## Pearson's Chi-squared test
##
## data:  tablica_zanr_award2
## X-squared = 16.803, df = 18, p-value = 0.5367
```

Udio pobjeda i nominacija po žanrovima

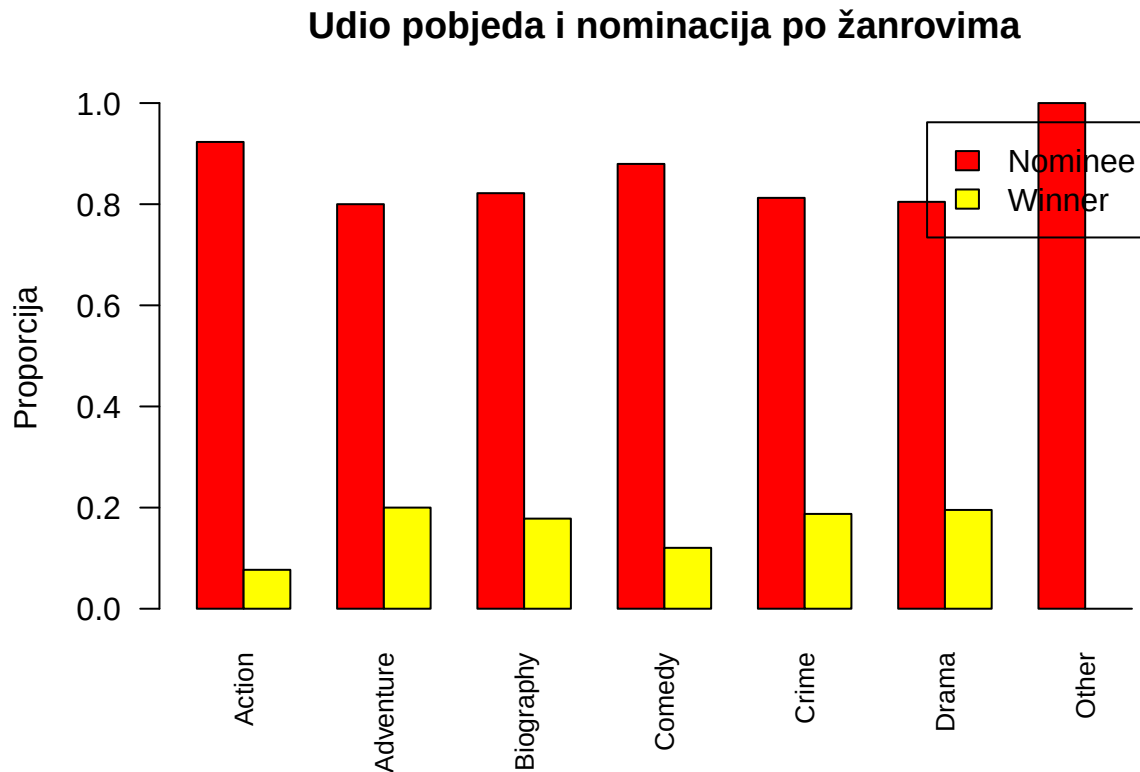


Udio pobjeda i nominacija po subžanrovima



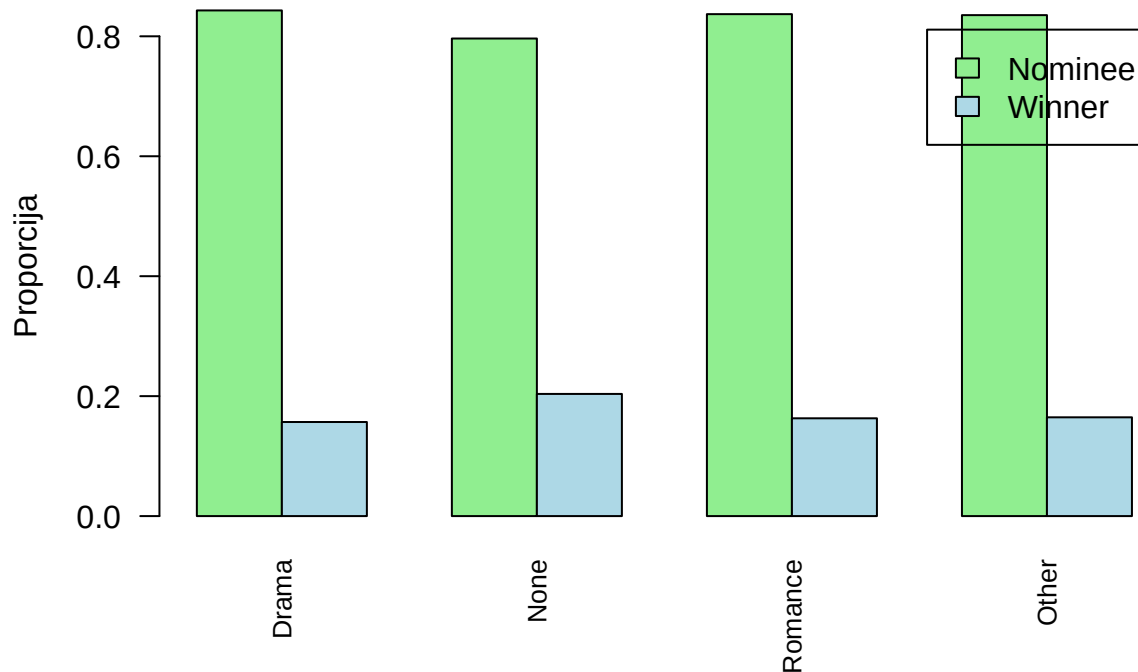
```
## [1] "Grupiranje rijetkih žanrova:"
## [1] "Nova kontingencijska tablica (grupirano):"
##
##      Nominee Winner
## Action      36      3
## Adventure   36      9
## Biography   83     18
## Comedy     95     13
## Crime       39      9
## Drama      173     42
## Other       15      0
## [1] "Očekivane frekvencije nakon grupiranja:"
##
##      Nominee  Winner
## Action  32.57968  6.420315
## Adventure 37.59194  7.408056
## Biography 84.37303 16.626970
## Comedy  90.22067 17.779335
## Crime   40.09807  7.901926
## Drama  179.60595 35.394046
## Other   12.53065  2.469352
##
## Pearson's Chi-squared test with simulated p-value (based on 2000
## replicates)
##
## data:  tablica_finalna
```

```
## X-squared = 8.8789, df = NA, p-value = 0.1659
```



```
## [1] "Grupiranje rijetkih subžanrova:"
## [1] "Subžanrovi s manje od 31 ukupnih filmova idu u 'Other'."
## [1] "Nova kontingencijska tablica (grupirano s pragom):"
##
##           Nominee Winner
## Drama         215     40
## None           43     11
## Romance        77     15
## Other         142     28
## [1] "Očekivane frekvencije nakon grupiranja (Sve bi trebale biti >= 5):"
##
##           Nominee  Winner
## Drama  213.02102 41.978984
## None   45.11033  8.889667
## Romance 76.85464 15.145359
## Other  142.01401 27.985989
##
## Pearson's Chi-squared test
##
## data:  tablica_finalna2
## X-squared = 0.71306, df = 3, p-value = 0.8701
```

Udio pobjeda po subžanrovima (Korigirano)



Na temelju originalnog testa H_0 odbacujemo, što znači da prihvaćamo H_1 , dok za test s grupiranim podacima Zadržavamo H_0 .

#Jesu li neki žanrovi u prosjeku dulji od ostalih? Hipoteze:

H_0 - Duljina filma ne ovisi o žanru filma H_1 - Duljina filma ovisi o žanru filma

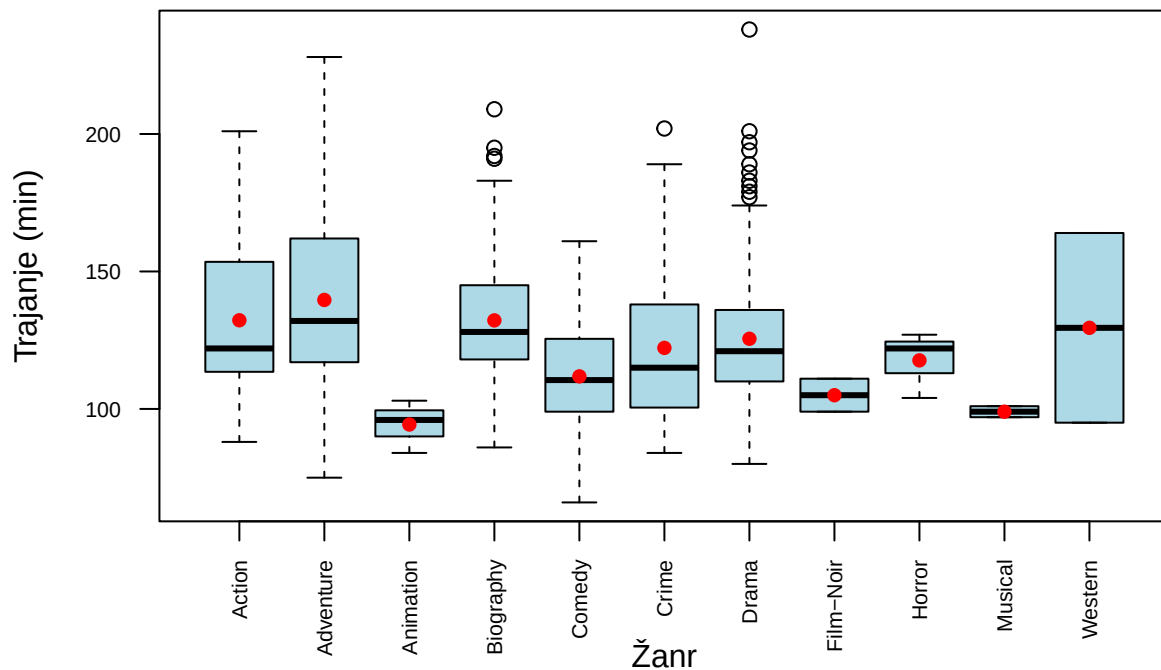
```
## [1] "Tablica duljine trajanja filmova u žanrovima:"
```

```
## # A tibble: 14 x 3
##   'Main Genre' Avg_Duration Movie_Count
##   <chr>         <dbl>         <int>
## 1 Adventure     140.           45
## 2 Action        132.           39
## 3 Biography     132.          101
## 4 Western       130.            2
## 5 Drama         126.          215
## 6 Crime         122.           48
## 7 Horror        118.            3
## 8 Family        115             1
## 9 Comedy        112.          108
## 10 Film-Noir     105             2
## 11 Musical        99             2
## 12 Animation     94.3            3
## 13 Music          84             1
## 14 Romance       84             1
```

```
## [1] "Tablica duljine trajanja filmova u žanrovima nakon filtriranja:"
```

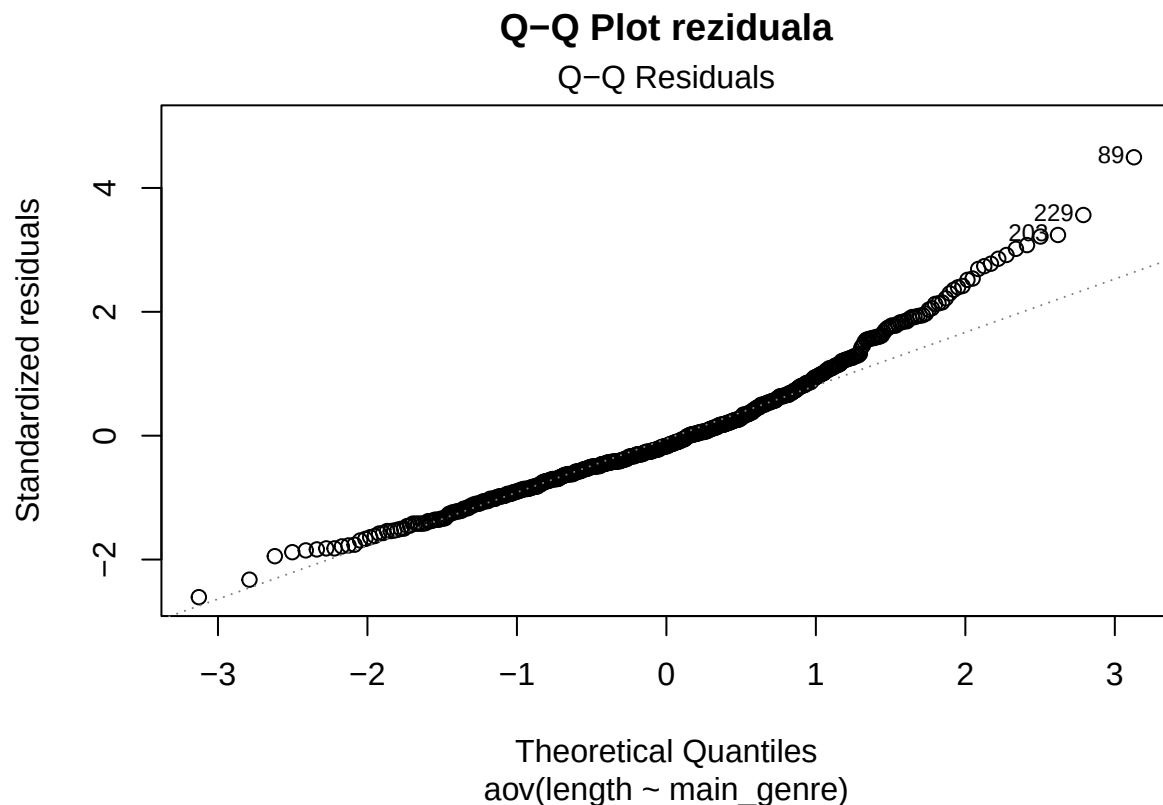
```
## # A tibble: 11 x 3
##   main_genre Avg_Duration Movie_Count
##   <fct>      <dbl>      <int>
## 1 Adventure    140.         45
## 2 Action       132.         39
## 3 Biography    132.        101
## 4 Western      130.          2
## 5 Drama        126.        215
## 6 Crime         122.         48
## 7 Horror        118.          3
## 8 Comedy        112.        108
## 9 Film-Noir     105          2
## 10 Musical       99          2
## 11 Animation     94.3         3
```

Trajanje filmova po žanrovima



```
##
## Shapiro-Wilk normality test
##
## data: residuals(model)
## W = 0.96046, p-value = 3.195e-11
```

```
# Q-Q Plot za normalnost
plot(model, which = 2, main = "Q-Q Plot reziduala")
```



Vizualni pregled Q-Q plota potvrđuje rezultate Shapiro-Wilk testa. Iako se većina podataka grupira oko teoretskog pravca normalnosti, postoje značajna odstupanja na krajevima distribucije. Ova odstupanja uzrokovana su ekstremnim vrijednostima (vrlo dugi filmovi), što je karakteristično za produkcije nominirane za nagradu Oscar. Zbog velikog uzorka, ova odstupanja ne ugrožavaju valjanost zaključka o razlikama među žanrovima

```
#Homogenost
print("Test homogenosti varijanca (Leveneov test):")
```

```
## [1] "Test homogenosti varijanca (Leveneov test):"
```

```
levene_res <- leveneTest(length ~ main_genre, data = data_clean)
print(levene_res)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
```

```
## group 10 2.8825 0.001613 **
##      557
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#ANOVA
print("ANOVA test:")
```

```
## [1] "ANOVA test:"
```

```
model <- aov(length ~ main_genre, data = data_clean)
levene_res <- leveneTest(length ~ main_genre, data = data_clean)

if(levene_res$`Pr(>F)`[1] > 0.05) {
  print("Standardni ANOVA test:") # Varijance su homogeneous -> Standardnu ANOVU
  print(summary(model))
} else {
  print("Welchov ANOVA test:") # Varijance nisu homogeneous (p < 0.05) -> Welchov ANOVA test
  welch_test <- oneway.test(length ~ main_genre, data = data_clean, var.equal = FALSE)
  print(welch_test)
}
```

```
## [1] "Welchov ANOVA test:"
##
## One-way analysis of means (not assuming equal variances)
##
## data: length and main_genre
## F = 13.214, num df = 10.000, denom df = 10.929, p-value = 9.722e-05
```

```
if(summary(model)[[1]][["Pr(>F)"]][1] < 0.05) {
  print("Post-hoc test (Tukey HSD):") # Usporedba parova žanrova
  tukey <- TukeyHSD(model)
  print(head(tukey$main_genre, 30)) # Prikaz prvih 30
}
```

```
## [1] "Post-hoc test (Tukey HSD):"
##
```

	diff	lwr	upr	p adj
Adventure-Action	7.38803419	-10.343156	25.1192245	9.600342e-01
Animation-Action	-37.92307692	-86.481942	10.6357877	2.926458e-01
Biography-Action	-0.03858847	-15.318043	15.2408656	1.000000e+00
Comedy-Action	-20.41381766	-35.554704	-5.2729315	7.938218e-04
Crime-Action	-10.04807692	-27.520113	7.4239590	7.438428e-01
Drama-Action	-6.73082886	-20.836772	7.3751144	9.048552e-01
Film-Noir-Action	-27.25641026	-86.016363	31.5035423	9.198480e-01
Horror-Action	-14.58974359	-63.148608	33.9691210	9.966446e-01
Musical-Action	-33.25641026	-92.016363	25.5035423	7.622404e-01
Western-Action	-2.75641026	-61.516363	56.0035423	1.000000e+00
Animation-Adventure	-45.31111111	-93.638190	3.0159676	8.910902e-02
Biography-Adventure	-7.42662266	-21.952631	7.0993858	8.586382e-01
Comedy-Adventure	-27.80185185	-42.182034	-13.4216696	4.550920e-08
Crime-Adventure	-17.43611111	-34.253223	-0.6189994	3.466023e-02
Drama-Adventure	-14.11886305	-27.404978	-0.8327484	2.649903e-02

```
## Film-Noir-Adventure -34.64444444 -93.212996 23.9241071 7.093750e-01
## Horror-Adventure -21.97777778 -70.304857 26.3493009 9.291301e-01
## Musical-Adventure -40.64444444 -99.212996 17.9241071 4.751518e-01
## Western-Adventure -10.14444444 -68.712996 48.4241071 9.999751e-01
## Biography-Animation 37.88448845 -9.597857 85.3668340 2.623794e-01
## Comedy-Animation 17.50925926 -29.928678 64.9471962 9.831372e-01
## Crime-Animation 27.87500000 -20.357598 76.1075975 7.380145e-01
## Drama-Animation 31.19224806 -15.925573 78.3100691 5.491535e-01
## Film-Noir-Animation 10.66666667 -63.318761 84.6520939 9.999956e-01
## Horror-Animation 23.33333333 -42.841245 89.5079112 9.880440e-01
## Musical-Animation 4.66666667 -69.318761 78.6520939 1.000000e+00
## Western-Animation 35.16666667 -38.818761 109.1520939 9.070343e-01
## Comedy-Biography -20.37522919 -31.593796 -9.1566621 4.089077e-07
## Crime-Biography -10.00948845 -24.218002 4.1990248 4.512904e-01
## Drama-Biography -6.69224039 -16.469124 3.0846430 4.969012e-01
```

P vrijednost u Leveneovom testu je $0.001613 < 0.05$ što znači da varijance nisu homogene što znači da trebamo koristiti Welchov ANOVA test.

U Welchovom ANOVA testu p-value = $9.722e-05$ što je znatno manje od 0.05 što znači da odbacujemo H_0 i zadržavamo H_1

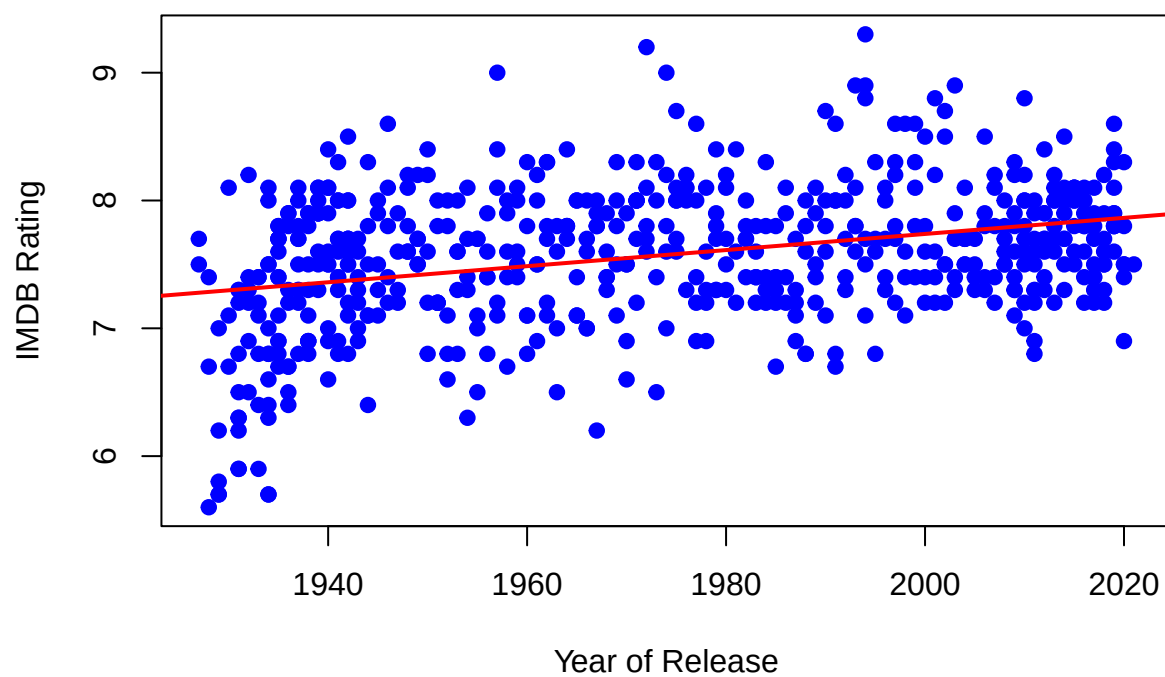
Promatrajući p adj stupac u Post-hoc Tuckey HSD testu vidimo na primjer da je komedija znatno kraća od ostalih žanrova (p adj < 0.05)

Kako su filmovi ocijenjeni s obzirom na njihovu starost?

```
mean.imdb.rating <- mean(oscars_dataset2$imdb_rating)
mean.tomato.rating <- oscars_dataset2$tomato_rating %>% na.omit() %>% mean()
mean.audience.rating <- oscars_dataset2$audience_rating %>% na.omit() %>% mean()
mean.year <- mean(oscars_dataset2$y_release)
imdb_ocjena_starost <- select(oscars_dataset2, y_release, imdb_rating)
```

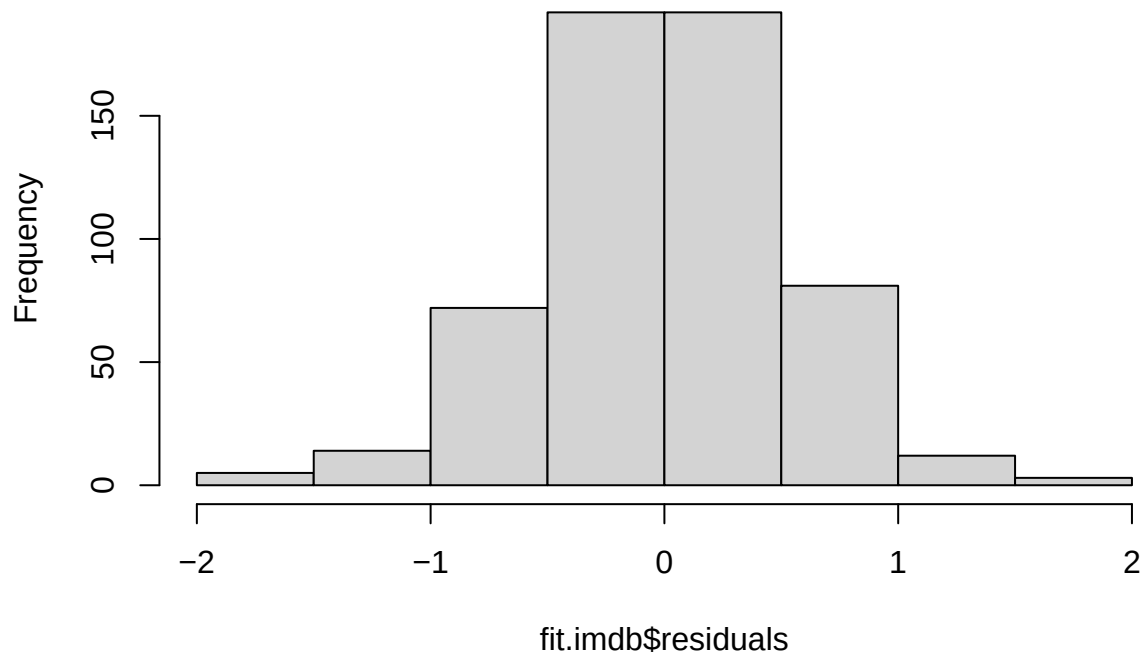
```
fit.imdb = lm(imdb_rating~y_release, data=imdb_ocjena_starost)
plot(imdb_ocjena_starost$y_release, imdb_ocjena_starost$imdb_rating, main = "IMDb Rating -Release Year",
     xlab = "Year of Release",
     ylab = "IMDb Rating",
     pch = 19,
     col = "blue")
#lines(imdb_ocjena_starost$y_release, fit.imdb$fitted.values, col='red')
abline(fit.imdb, col = "red", lwd = 2)
```

IMDb Rating –Release Year



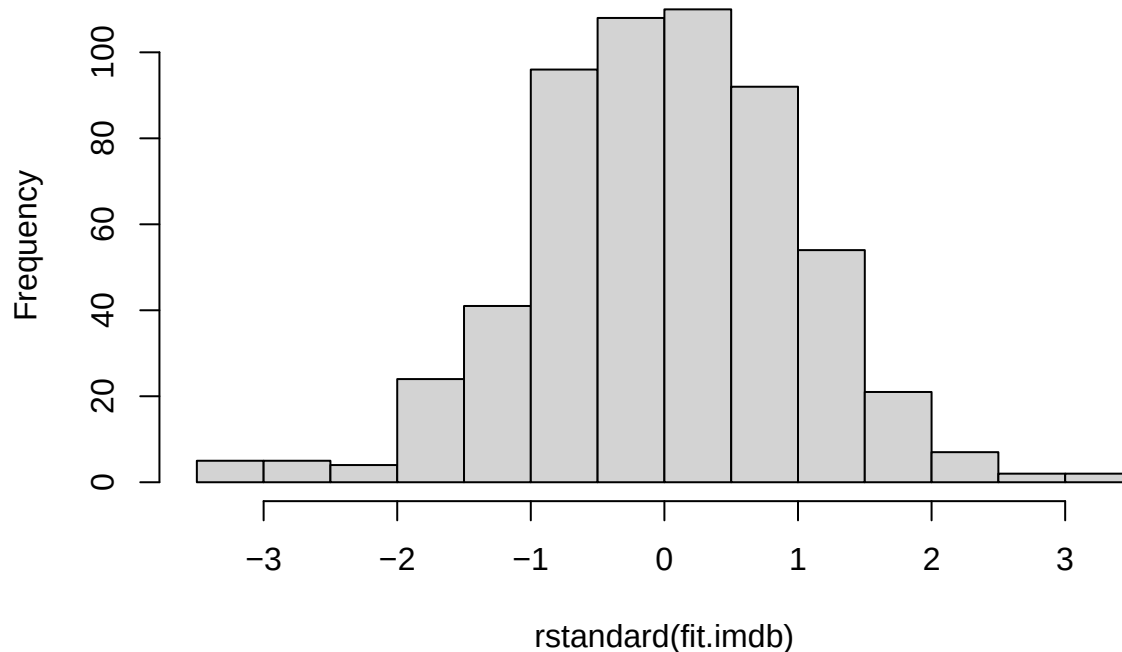
```
#raspodjela reziduala  
hist(fit.imdb$residuals)
```

Histogram of fit.imdb\$residuals



```
#studentova standardizacija reziduala  
hist(rstandard(fit.imdb))
```

Histogram of rstandard(fit.imdb)



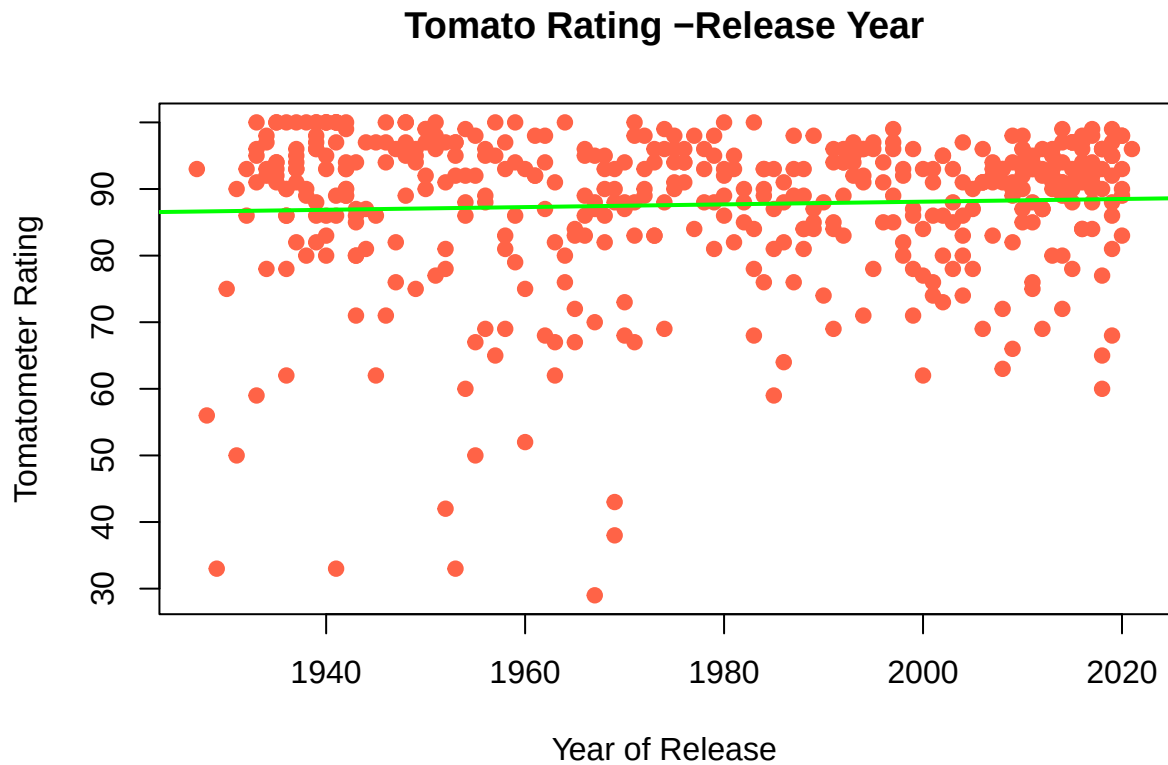
```
#iscitavanje podataka uključujući prilagođeni R^2
summary(fit.imdb)
```

```
##
## Call:
## lm(formula = imdb_rating ~ y_release, data = imdb_ocjena_starost)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.68474 -0.35157  0.00511  0.36096  1.63815
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.8580725   1.4910503   -3.258  0.00119 **
## y_release    0.0062981   0.0007555    8.336 5.78e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5288 on 569 degrees of freedom
## Multiple R-squared:  0.1088, Adjusted R-squared:  0.1073
## F-statistic: 69.49 on 1 and 569 DF,  p-value: 5.783e-16
```

```
#korelacije
cor(oscars_dataset2$y_release,oscars_dataset2$imdb_rating)
```

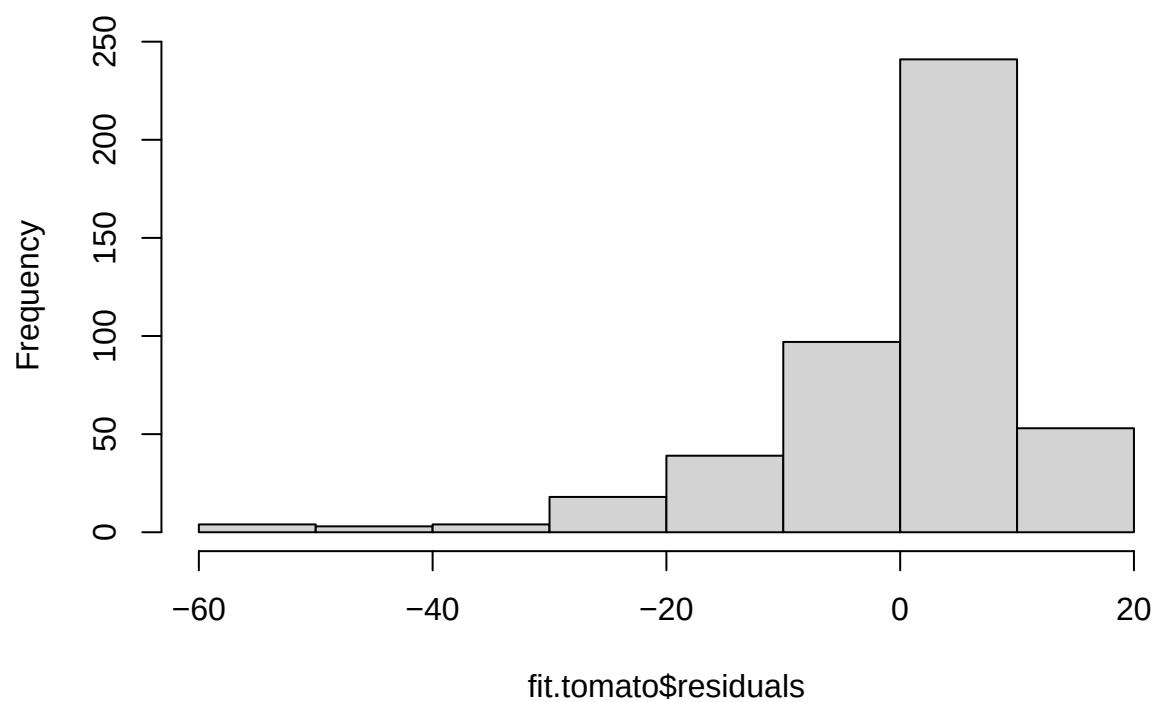
```
## [1] 0.3299097
```

```
fit.tomato = lm(tomato_rating~y_release,data=oscars_dataset2)
plot(oscars_dataset2$y_release,oscars_dataset2$tomato_rating,main = "Tomato Rating -Release Year",
     xlab = "Year of Release",
     ylab = "Tomatometer Rating",
     pch = 19,
     col = "tomato")
abline(fit.tomato, col = "green", lwd = 2)
```



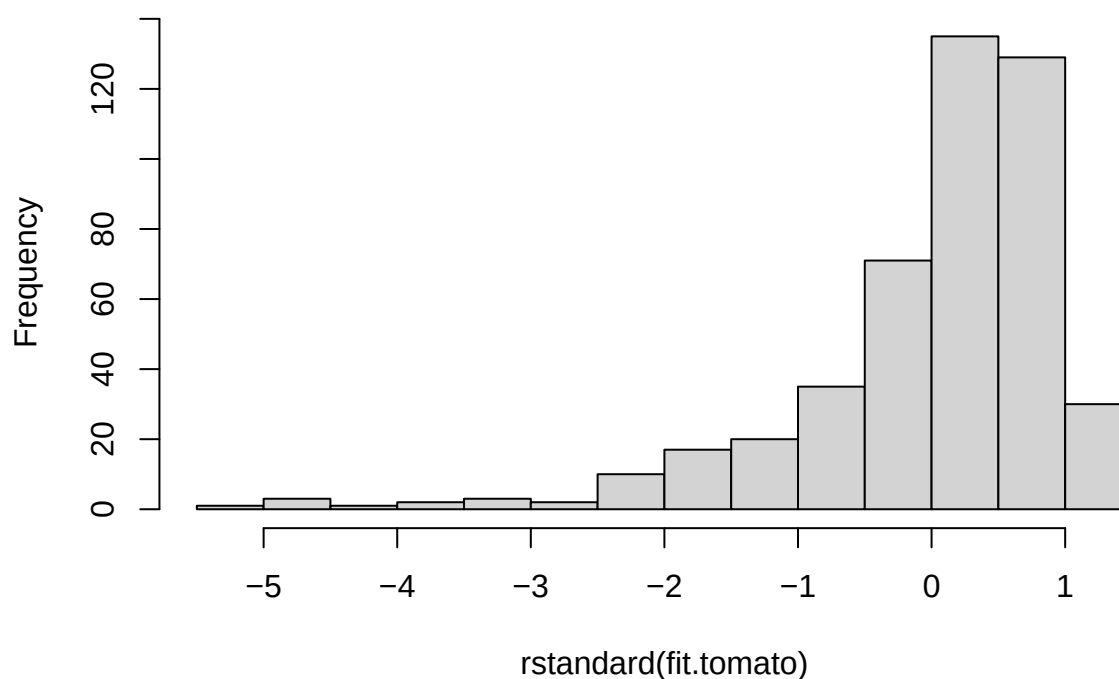
```
hist(fit.tomato$residuals)
```

Histogram of fit.tomato\$residuals



```
hist(rstandard(fit.tomato))
```

Histogram of rstandard(fit.tomato)

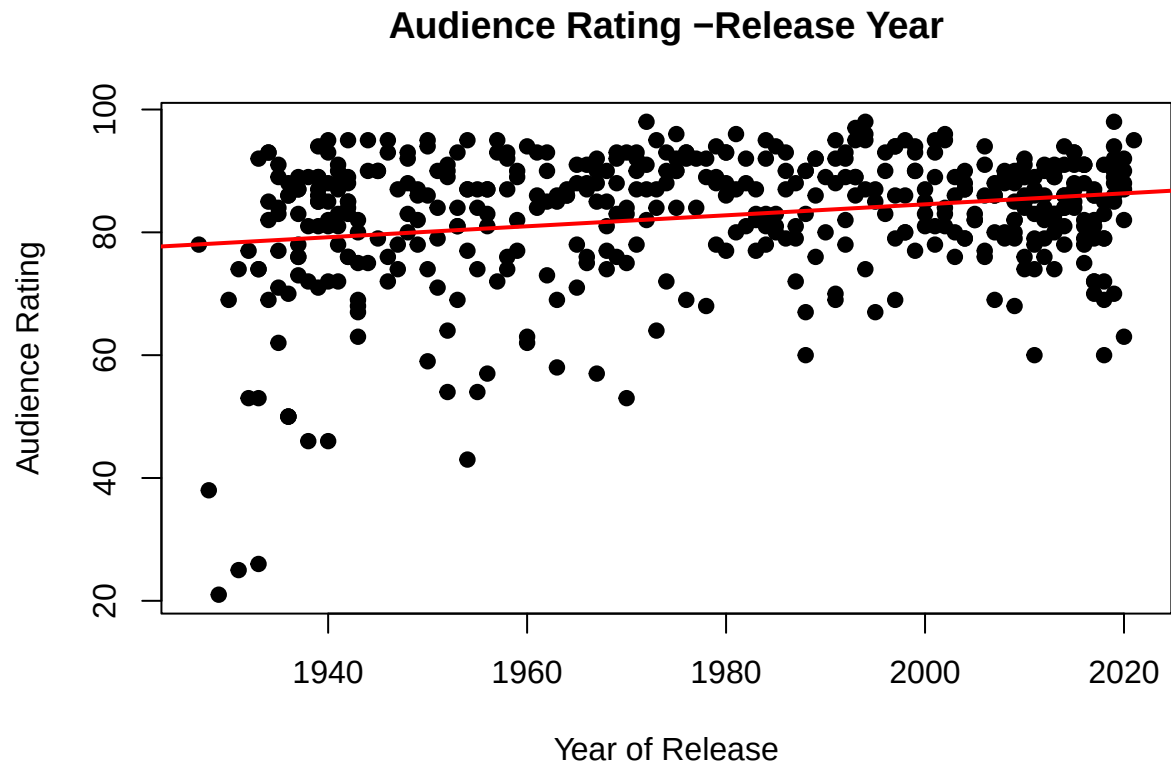


```
summary(fit.tomato)
```

```
##
## Call:
## lm(formula = tomato_rating ~ y_release, data = oscars_dataset2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -58.427  -3.897   3.079   7.546  13.262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  47.55938   37.29628   1.275   0.203
## y_release     0.02027    0.01886   1.075   0.283
##
## Residual standard error: 11.53 on 457 degrees of freedom
## (112 observations deleted due to missingness)
## Multiple R-squared:  0.002522,    Adjusted R-squared:  0.0003392
## F-statistic: 1.155 on 1 and 457 DF,  p-value: 0.283
```

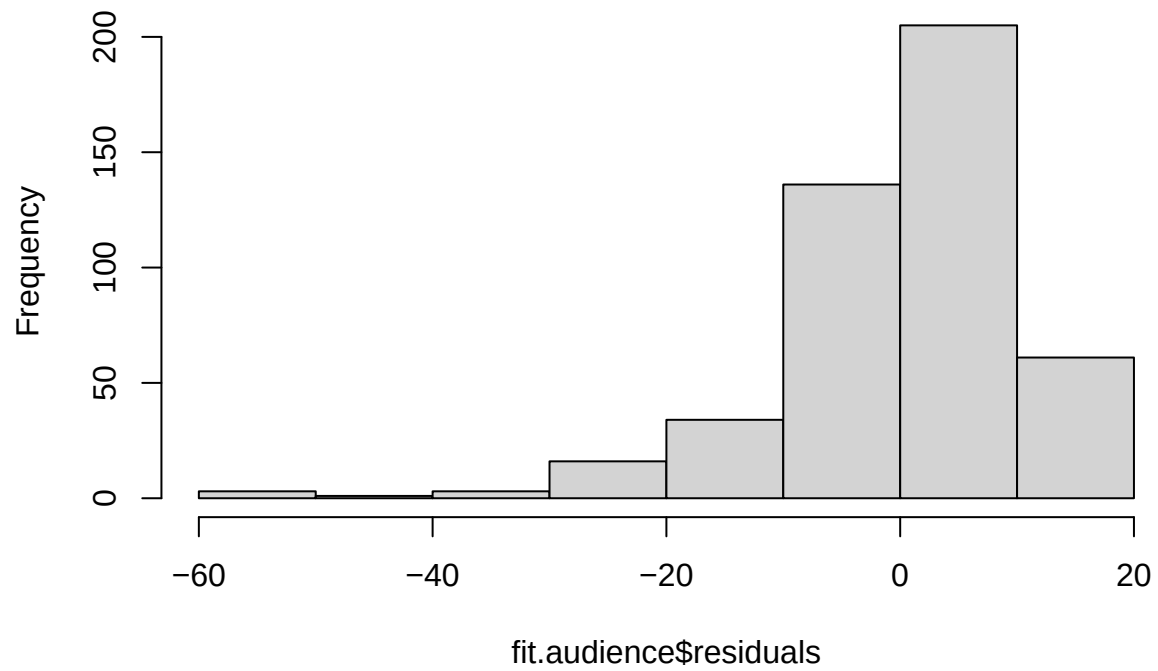
```
fit.audience = lm(audience_rating~y_release,data=oscars_dataset2)
plot(oscars_dataset2$y_release,oscars_dataset2$audience_rating,main = "Audience Rating -Release Year",
     xlab = "Year of Release",
     ylab = "Audience Rating",
     pch = 19,
```

```
col = "black")  
abline(fit.audience, col = "red", lwd = 2)
```



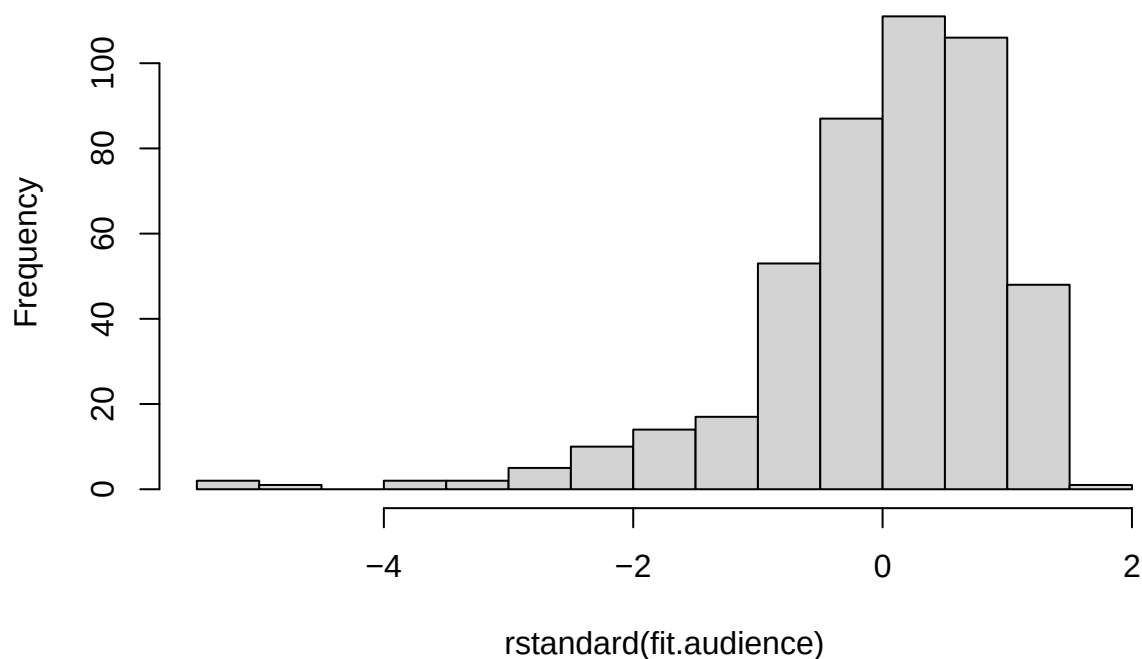
```
hist(fit.audience$residuals)
```

Histogram of fit.audience\$residuals



```
hist(rstandard(fit.audience))
```

Histogram of rstandard(fit.audience)



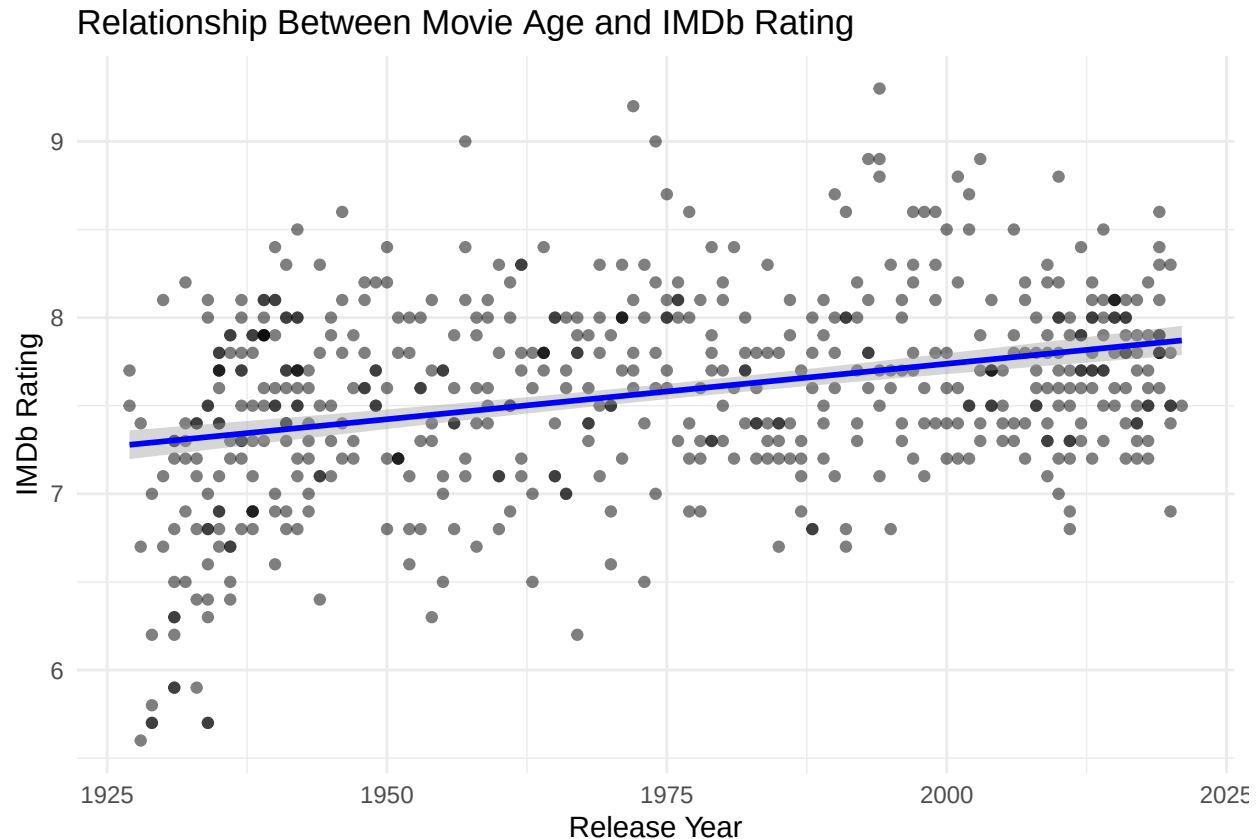
```
summary(fit.audience)
```

```
##
## Call:
## lm(formula = audience_rating ~ y_release, data = oscars_dataset2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -57.218  -4.694   2.008   7.097  15.940
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -94.14921   34.26520  -2.748  0.00624 **
## y_release     0.08936    0.01732   5.158 3.73e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.59 on 457 degrees of freedom
## (112 observations deleted due to missingness)
## Multiple R-squared:  0.05501,    Adjusted R-squared:  0.05295
## F-statistic: 26.61 on 1 and 457 DF,  p-value: 3.726e-07
```

Očitujemo malu vrijednost R^2 čime dalje potvrđujemo tvrdnju da starost filma nema utjecaj na ocjenu. Za imdb je R^2

```
ggplot(imdb_ocjena_starost, aes(x = y_release, y = imdb_rating)) +
  geom_point(alpha = 0.5) + # alpha makes points semi-transparent to see density
  geom_smooth(method = "lm", color = "blue", se = TRUE) + # "lm" = linear model
  labs(title = "Relationship Between Movie Age and IMDb Rating",
        x = "Release Year",
        y = "IMDb Rating") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



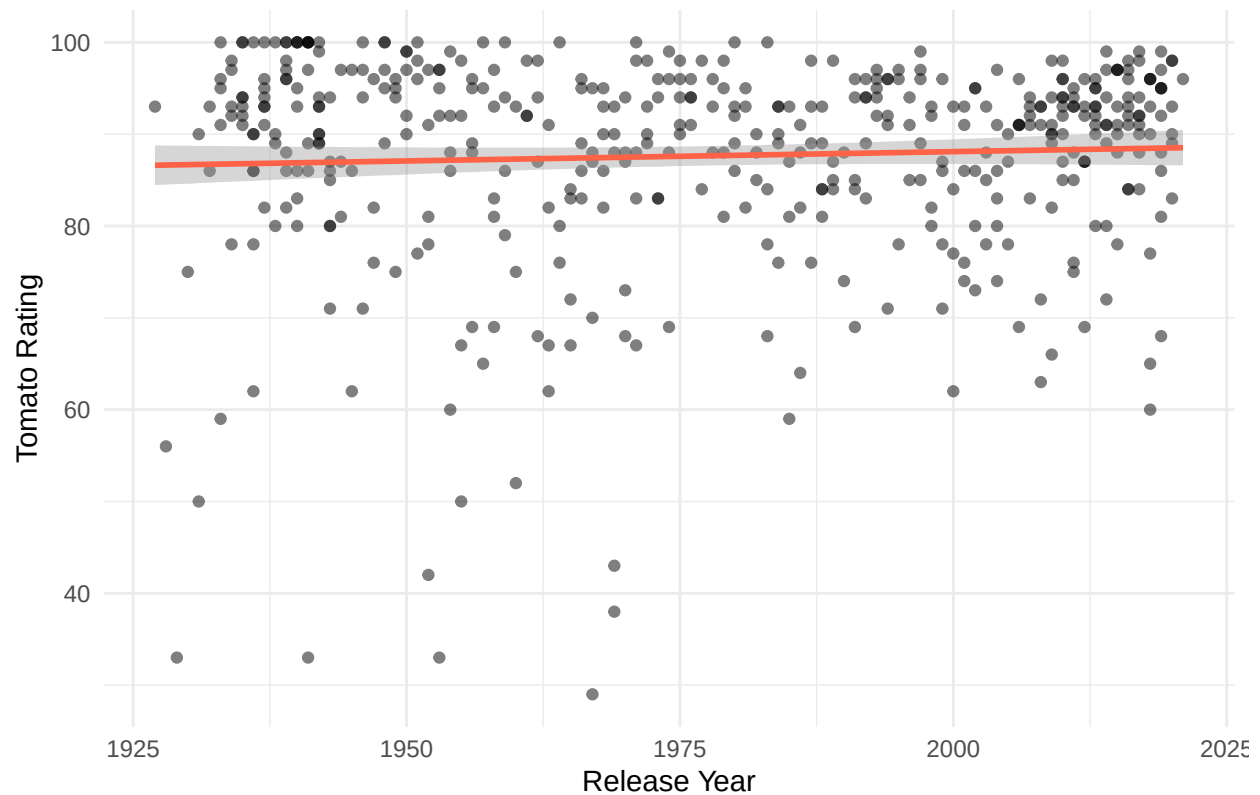
```
ggplot(oscars_dataset2, aes(x = y_release, y = tomato_rating)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "tomato", se = TRUE) +
  labs(title = "Relationship Between Movie Age and Tomatometer Rating",
        x = "Release Year",
        y = "Tomato Rating") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 112 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 112 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Relationship Between Movie Age and Tomatometer Rating



```
ggplot(oscars_dataset2, aes(x = y_release, y = audience_rating)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", color = "green", se = TRUE) +  
  labs(title = "Relationship Between Movie Age and Audience Rating",  
        x = "Release Year",  
        y = "Audience Rating") +  
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

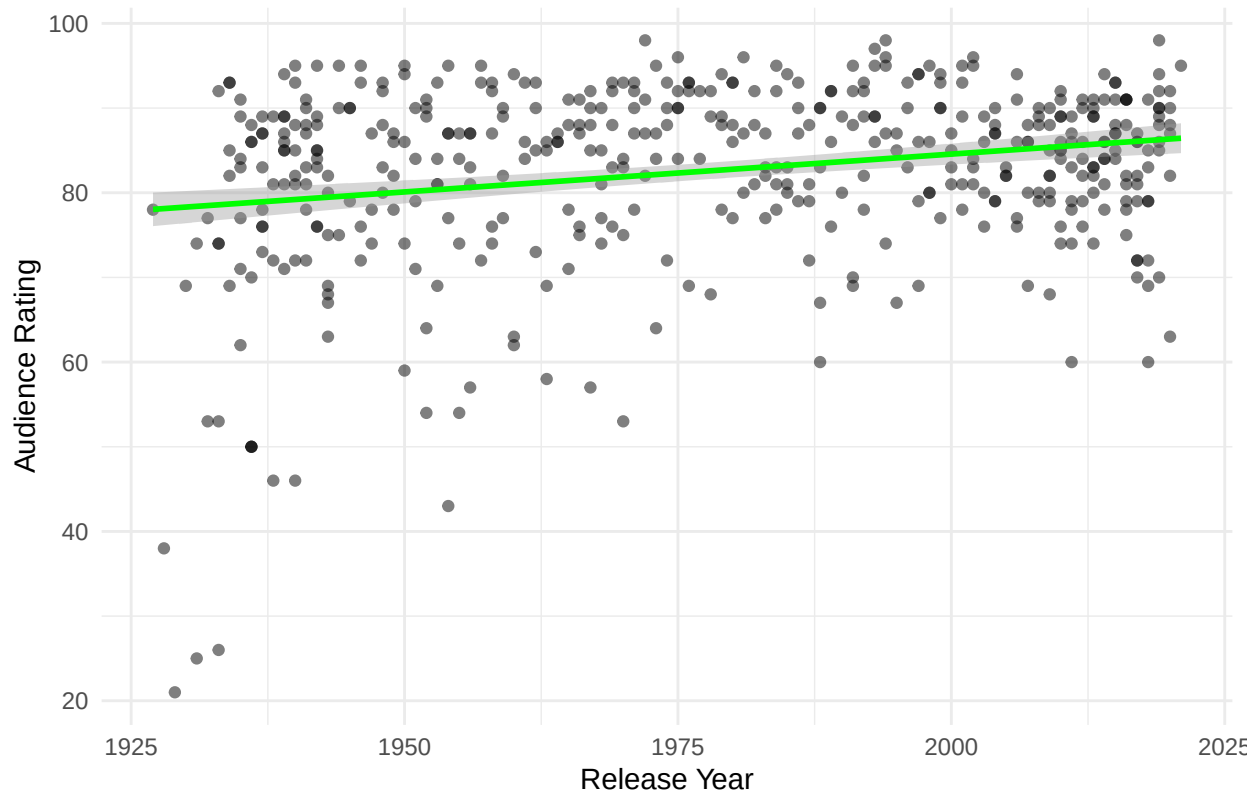
```
## Warning: Removed 112 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Removed 112 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```

Relationship Between Movie Age and Audience Rating



Moguće je na prvi pogled zaključiti kako godina izdanja ne utječe mnogo na ocjenu filma. Mali pozitivan nagib je upečaćen no ne smatramo ga toliko signifikantnim. Najveći nagib je u IMDB grafu.

```
# test normalnosti reziduala  $p < 0.05$   
shapiro.test(fit.imdb$residuals)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: fit.imdb$residuals  
## W = 0.99448, p-value = 0.03708
```

```
shapiro.test(fit.tomato$residuals)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: fit.tomato$residuals  
## W = 0.82161, p-value < 2.2e-16
```

```
shapiro.test(fit.audience$residuals)
```

```
##  
## Shapiro-Wilk normality test
```

```
##  
## data: fit.audience$residuals  
## W = 0.88293, p-value < 2.2e-16
```

p vrijednost je manje od 0.05 za sve a možemo zaključiti normalnost

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$imdb_rating,method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: oscars_dataset2$y_release and oscars_dataset2$imdb_rating  
## t = 8.3363, df = 569, p-value = 5.783e-16  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.2547525 0.4011050  
## sample estimates:  
## cor  
## 0.3299097
```

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$tomato_rating,method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: oscars_dataset2$y_release and oscars_dataset2$tomato_rating  
## t = 1.0749, df = 457, p-value = 0.283  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## -0.04149945 0.14109647  
## sample estimates:  
## cor  
## 0.05021814
```

```
cor.test(oscars_dataset2$y_release, oscars_dataset2$audience_rating,method = "pearson")
```

```
##  
## Pearson's product-moment correlation  
##  
## data: oscars_dataset2$y_release and oscars_dataset2$audience_rating  
## t = 5.158, df = 457, p-value = 3.726e-07  
## alternative hypothesis: true correlation is not equal to 0  
## 95 percent confidence interval:  
## 0.1461621 0.3192250  
## sample estimates:  
## cor  
## 0.2345512
```

Nemožemo odbaciti hipotezu da su reziduali normalno distribuirani. Pearson korelacijskim testom možemo vidjeti da su generalno slabo korelirani stime da imdb ima najveću korelaciju sa 0.33 što se može smatrati umjerenom korelacijom. Kako su sve tri umjerene no pozitivne postoji mogućnost da noviji filmovi imaju bolje ocijene.

Možemo li naslutiti kako je film ocijenjen pomoću drugih značajki?

Možemo iskoristiti nekoliko metoda kako bi vidjeli možemo li koristiti druge merike da bi mogli naslutiti ocjenu filma.

Prvo krećemo sa numeričkom analizom.

```
# Biramo samo numeričke stupce za nalizu
numerical_data <- oscars_dataset2[, c("length", "imdb_rating", "imdb_votes",
                                       "tomato_rating", "tomato_count", "audience_rating", "audience_count")]

# Kalkuliramo matrix korelacije koji nam je pogod
cor_matrix <- cor(numerical_data, use = "complete.obs")
print(cor_matrix)
```

```
##           length imdb_rating imdb_votes tomato_rating tomato_count
## length      1.00000000  0.2334096 0.26604443  -0.18968511  0.07762022
## imdb_rating  0.23340962  1.0000000 0.61672439   0.46632079  0.24633971
## imdb_votes   0.26604443  0.6167244 1.00000000   0.09744051  0.47035921
## tomato_rating -0.18968511  0.4663208 0.09744051   1.00000000  0.12644534
## tomato_count  0.07762022  0.2463397 0.47035921   0.12644534  1.00000000
## audience_rating 0.13647250  0.8485251 0.33591478   0.50462680  0.15622655
## audience_count 0.18911348  0.1679893 0.31230476  -0.00915894  0.04615373
##           audience_rating audience_count
## length      0.13647250      0.18911348
## imdb_rating  0.84852507      0.16798930
## imdb_votes   0.33591478      0.31230476
## tomato_rating 0.50462680     -0.00915894
## tomato_count  0.15622655      0.04615373
## audience_rating 1.00000000     -0.01560293
## audience_count -0.01560293      1.00000000
```

Ne primjećujemo znatne korelacije koje bi nam mogle korisiti u predikciji

Moramo analizirati kvalitativne stupce.

```
genre_analysis <- oscars_dataset2 %>%
  group_by(main_genre) %>%
  summarise(
    avg_rating = mean(imdb_rating, na.rm = TRUE),
    movie_count = n()
  ) %>%
  arrange(desc(avg_rating))

print(genre_analysis)
```

```
## # A tibble: 14 x 3
##   main_genre avg_rating movie_count
##   <chr>      <dbl>      <int>
## 1 Animation      8.13          3
## 2 Crime          7.88         48
## 3 Family         7.8           1
```

##	4	Horror	7.77	3
##	5	Action	7.65	39
##	6	Biography	7.58	101
##	7	Drama	7.56	215
##	8	Adventure	7.56	45
##	9	Comedy	7.47	108
##	10	Film-Noir	7.45	2
##	11	Romance	6.7	1
##	12	Western	6.35	2
##	13	Musical	6.25	2
##	14	Music	5.7	1

Možemo se i zapitati imaju li Oscar pobjednici bolje ocjene.

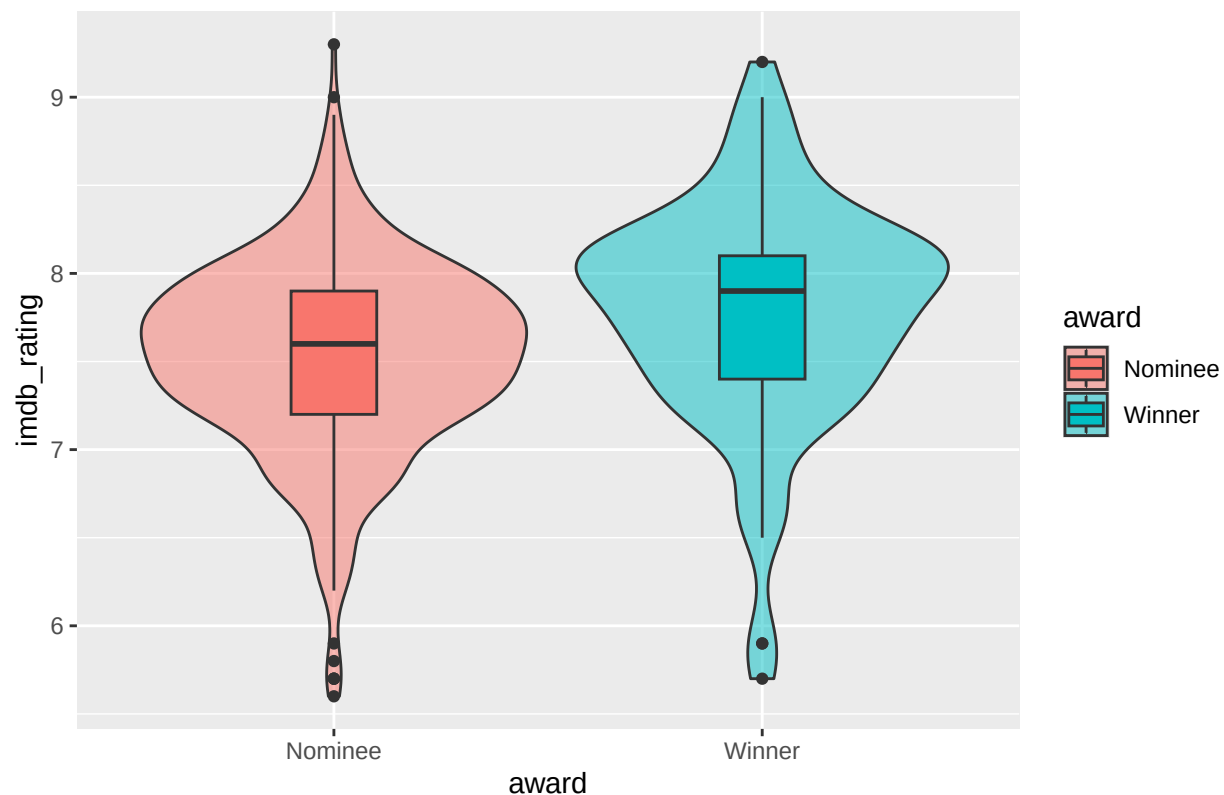
```
imdb_stats <- oscars_dataset %>%
  group_by(Award) %>%
  summarise(
    Prosjek_IMDb = mean(`IMDB Rating`, na.rm = TRUE),
    Medijan_IMDb = median(`IMDB Rating`, na.rm = TRUE),
    Broj_filmova = n()
  )

print(imdb_stats)
```

```
## # A tibble: 2 x 4
##   Award   Prosjek_IMDb Medijan_IMDb Broj_filmova
##   <chr>       <dbl>       <dbl>       <int>
## 1 Nominee     7.53         7.6         477
## 2 Winner      7.78         7.9         94
```

```
#Vidimo da ima vidljive razlike u prosjeku i u medijanu
ggplot(oscars_dataset2, aes(x = award, y = imdb_rating, fill = award)) +
  geom_violin(alpha = 0.5) +
  geom_boxplot(width = 0.2) +
  labs(title = "Raspodjela ocjena oscan nimiracija i pobjednika?")
```

Raspodjela ocjena oscan niminacija i pobjednika?



```
# Pomoću t testa možemo testirati jesu li prosjeci ova dva skupa stvarno statistički različita
# Znamo da su dva skupa nezavisna ner ne dijele istu značajku
#Koristi ćemo t test pa prvo se moramo uvjeriti normalnost podataka"
winners <- oscars_dataset2 %>% filter(award == "Winner") %>% select(imdb_rating) %>% pull(imdb_rating)
shapiro.test(winners)
```

```
##
## Shapiro-Wilk normality test
##
## data: winners
## W = 0.95032, p-value = 0.001322
```

```
nominees <- oscars_dataset2 %>% filter(award == "Nominee") %>% select(imdb_rating) %>% pull(imdb_rating)
shapiro.test(nominees)
```

```
##
## Shapiro-Wilk normality test
##
## data: nominees
## W = 0.97874, p-value = 1.939e-06
```

```
# pošto je  $p < 0.01$  za oba testa možemo zaključiti da nisu normalno distribuirani
#Moramo koristiti ne parametarsku metodu
wilcox.test(winners, nominees, alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: winners and nominees
## W = 28913, p-value = 4.294e-06
## alternative hypothesis: true location shift is greater than 0
```

p < 0.0001 možemo gotovo sigurno odbaciti hipotezu u korist toga da oscar pobjednici imaju bolji rating

Biti pobjednik oscara jedan je od prediktora višeg ratinga

Možemo pretpostaviti da neki producenti proizvode bolje filmove

*#Kako bi mogli vidjeti koji studiji proizvode visoko ocjenjene filmove moramo znati interval pouzdanosti
#Koristimo bootstrap metodu jer imdb rating nisu normalno distribuirani i robusno je*

Funkcija koja računa prosjek (potrebna za bootstrap)

```
boot_mean <- function(data, indices) {
  return(mean(data[indices], na.rm = TRUE))
}
```

Pokretanje bootstrapa (1000 simulacija)

set.seed(553343) # Za ponovljivost

```
boot_results <- boot(data = oscars_dataset2$imdb_rating, statistic = boot_mean, R = 1000)
```

Dobivanje 95% intervala pouzdanosti

```
ci <- boot.ci(boot_results, conf = 0.99, type = "perc")
```

#perc type Directly uses quantiles from the bootstrap distribution as interval endpoints (e.g., the 2.5

#types valjda treba ili perc ili basic

```
gornja_granica <- ci$percent[5]
```

#Sve produkcijske kompanije koje su iznad gornje granice možemo poprilično sigurno reći da proizvode visoko

```
uspjesni_producenti <- oscars_dataset2 %>%
```

```
  group_by(producer) %>%
```

```
  summarise(
```

```
    prosjek_producenta = mean(imdb_rating, na.rm = TRUE),
```

```
    broj_filmova = n()
```

```
  ) %>%
```

```
  filter(broj_filmova >= 3) %>%
```

```
  filter(prosjek_producenta > gornja_granica) %>%
```

```
  arrange(desc(prosjek_producenta))
```

```
print(uspjesni_producenti)
```

```
## # A tibble: 7 x 3
```

```
##   producer
```

```
##   <chr>
```

```
## 1 Lawrence Bender
```

```
## 2 Stanley Kubrick
```

```
## 3 Irwin Winkler and Robert Chartoff
```

```
## 4 Sam Spiegel
```

```
## 5 Stanley Kramer
```

```
## 6 Robert Wise
```

```
prosijek_producenta broj_filmova
```

```
<dbl>
```

```
<int>
```

```
8.5
```

```
3
```

```
8.27
```

```
3
```

```
8.03
```

```
3
```

```
7.92
```

```
4
```

```
7.73
```

```
6
```

```
7.7
```

```
3
```

Zaključujemo da su ovih 7 producenata proizvode statistički bolje filmove po tome što imaju više filmova (znači da imamo barem neku populaciju za proučiti) i prosječna ocjena njihovih filmova je veća od gornje granice intervala pouzdanosti za $\alpha = 0.01$

#Možemo li pomoću dostupnih značajki pouzdano razlikovati pobjednike od ostalih?

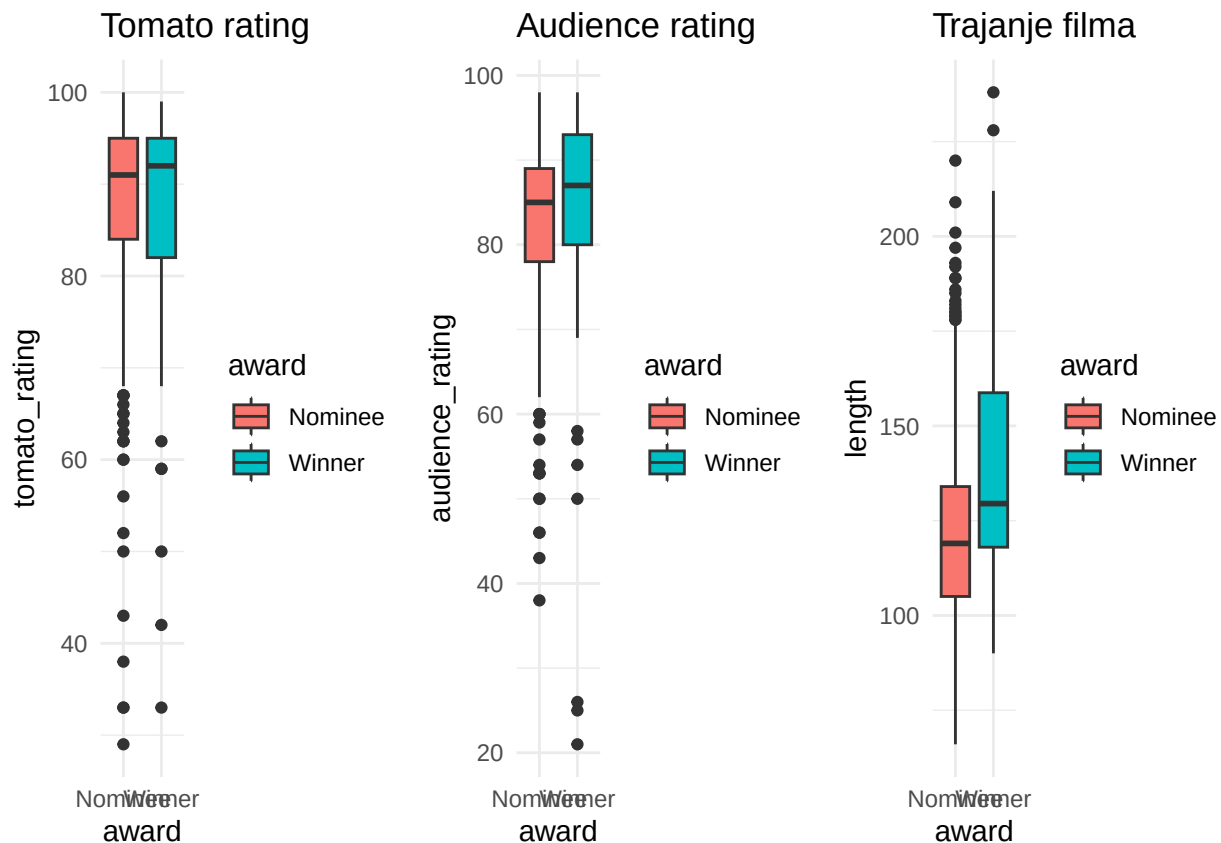
```
# 1. Vizualizacija - Boxplotovi za ključne metrike
p1 <- ggplot(oscars_dataset2, aes(x = award, y = tomato_rating, fill = award)) +
  geom_boxplot() +
  labs(title = "Tomato rating") + theme_minimal()

p2 <- ggplot(oscars_dataset2, aes(x = award, y = length, fill = award)) +
  geom_boxplot() +
  labs(title = "Trajanje filma") + theme_minimal()

p3 <- ggplot(oscars_dataset2, aes(x = award, y = audience_rating, fill = award)) +
  geom_boxplot() +
  labs(title = "Audience rating") + theme_minimal()

grid.arrange(p1, p3, p2, ncol = 3)
```

```
## Warning: Removed 112 rows containing non-finite outside the scale range
## ('stat_boxplot()').
## Removed 112 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
# 2. Statistički testovi (Wilcoxon rank-sum test)
# H0: Nema razlike u distribuciji između pobjednika i nominiranih
test_imdb <- wilcox.test(imdb_rating ~ award, data = oscars_dataset2)
test_len <- wilcox.test(length ~ award, data = oscars_dataset2)
test_tomato <- wilcox.test(tomato_rating ~ award, data = oscars_dataset2)

print(paste("P-vrijednost za IMDb:", test_imdb$p.value))
```

```
## [1] "P-vrijednost za IMDb: 8.58788614504331e-06"
```

```
print(paste("P-vrijednost za Duljinu:", test_len$p.value))
```

```
## [1] "P-vrijednost za Duljinu: 6.71961208650246e-06"
```

```
print(paste("P-vrijednost za Tomato:", test_tomato$p.value))
```

```
## [1] "P-vrijednost za Tomato: 0.91513884267567"
```